SEC Comment Letter Radar: Final Results Brief

This brief summarizes the two final-results sections of the Streamlit dashboard for team communication:

- Topic CAR
- Regression Playground

The goal is to explain what each page shows, what the main numeric results mean, and how the results answer the project research question.

Research Question

Do SEC comment letters contain regulatory signals that are reflected in stock returns, and do topic and severity measures explain variation in market reaction?

Hypotheses

Hypothesis	Question
H1: Market reaction	Do SEC comment-letter events have abnormal stock-return reactions around the event date?
H2: Post-event drift	Do abnormal returns continue after the immediate event window?
H3: Topic heterogeneity	Do different comment-letter topics produce different CAR patterns?
H4: Severity effect	Are more severe comment letters associated with stronger CAR reactions?

1. Topic CAR Page

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Final Results

Official event-study and regression outputs from the current project pipeline.

Events	Regression included	Topic groups	Industries
283	273	6	11

Topic CAR Regression Playground

Topic CAR

This view compares average cumulative abnormal returns across topic groups.

Topic CAR Summary

topic	n_events	mean_CAR_-3_-1	mean_CAR_-1_3	mean_CAR_1_60	mean_severity
accounting_financial_reporting	132	0.40%	-0.53%	0.10%	1.0124
business_structure	22	0.28%	-0.02%	-5.36%	1.4592
mda_liquidity	40	-0.78%	0.25%	-0.32%	-0.1441
revenue_non_gaap	31	0.49%	1.01%	1.24%	0.1045
risk_legal_regulatory	10	-1.53%	0.57%	-5.75%	1.701
routine_other	48	-0.09%	-0.40%	0.61%	-1.5255

Topic

All topics

Relative trading-day window

-10 60

What This Page Shows

The Topic CAR page is the event-study result page. It groups events by the V3 topic classifier and compares average cumulative abnormal returns across topics and event windows.

The main table reports topic-level mean CAR for three windows:

Window	Interpretation
`mean_CAR_-3_-1`	Pre-event movement before the SEC comment-letter event
`mean_CAR_-1_3`	Immediate market reaction around the event date
`mean_CAR_1_60`	Post-event drift after the event

The line charts show average CAR over relative trading days. The shaded windows help connect the visual event-study curve to the three CAR windows used in the summary table.

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> Topic CAR Code Logic

> Static Figure Snapshots

Topic CAR Conclusions

- **Market reaction exists, but it is mixed.** Immediate CAR ranges from `accounting_financial_reporting` (-0.53%) to `revenue_non_gaap` (1.01%). This gives mixed support for H1, not a clean negative-reaction story.
- **Topic heterogeneity is visible.** Different topics produce different CAR patterns, which supports the descriptive part of H3.
- **Post-event drift exists for some topics.** Drift ranges from `risk_legal_regulatory` (-5.75%) to `revenue_non_gaap` (1.24%). This gives partial support for H2.
- **Overall conclusion:** Topic CAR suggests SEC comment letters contain market-relevant signals, but the direction and strength depend on content.

Main Numeric Results

Result	Value
Lowest immediate CAR, `CAR_−1_3`	`accounting_financial_reporting`: -0.53%
Highest immediate CAR, `CAR_−1_3`	`revenue_non_gaap`: +1.01%
Lowest post-event drift, `CAR_1_60`	`risk_legal_regulatory`: -5.75%
Highest post-event drift, `CAR_1_60`	`revenue_non_gaap`: +1.24%

Topic CAR Interpretation

- Market reaction exists, but it is mixed rather than uniformly negative.
- Topic heterogeneity is visible: different comment-letter topics show different CAR patterns.
- The strongest descriptive post-event drift is negative for `risk_legal_regulatory` and positive for `revenue_non_gaap`.

- Topic CAR provides descriptive support for H2 and H3, while H1 is only partially supported because the direction is not uniformly negative.

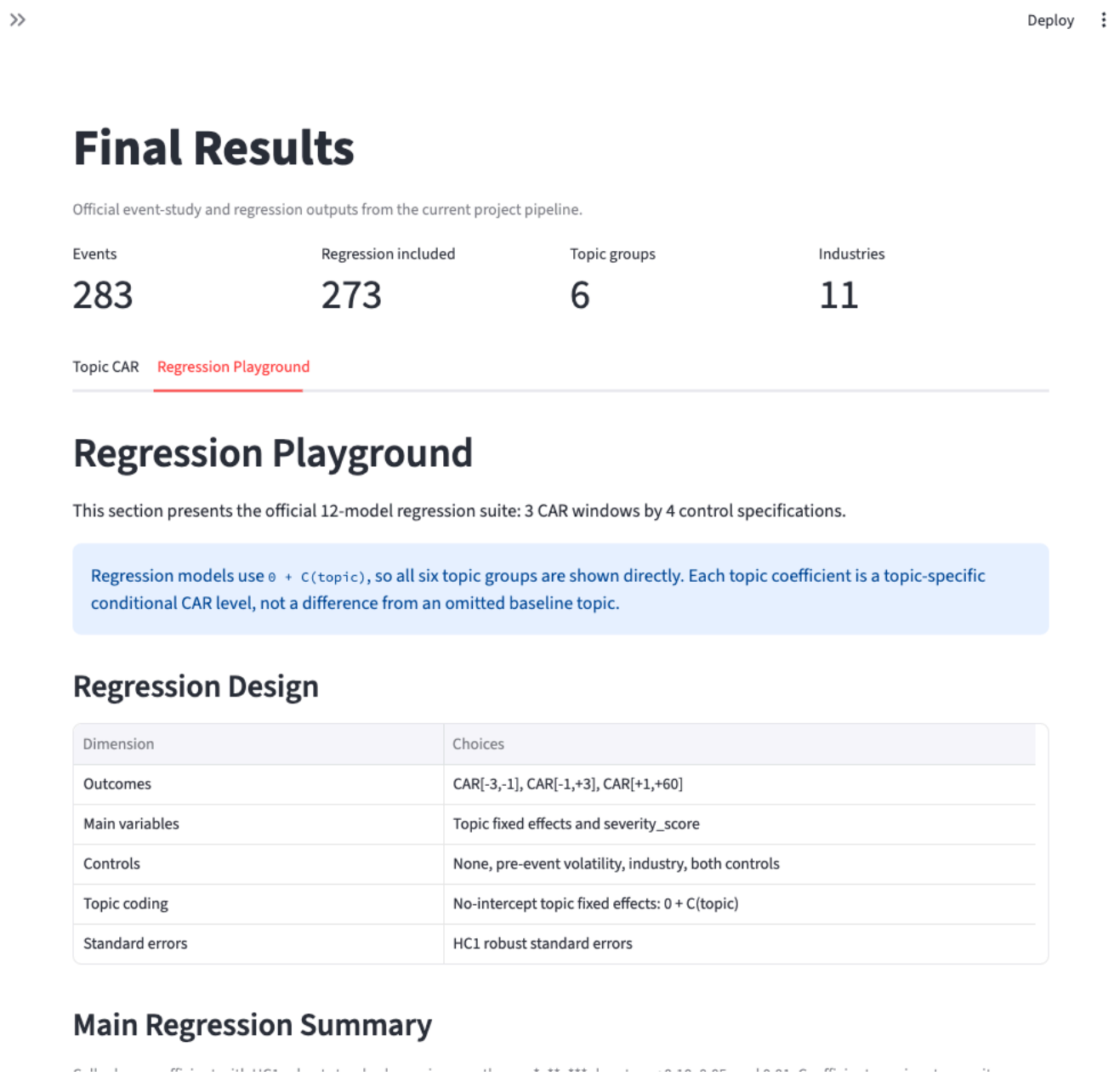
How to Explain This to Teammates

Topic CAR answers the descriptive event-study question:

> Around SEC comment-letter events, do returns move differently depending on what the letter is about?

The answer from the current baseline sample is yes, but the pattern is not one simple negative effect. Topic matters descriptively, and the strongest visual evidence is that market reaction differs across letter content categories.

2. Regression Playground Page



What This Page Shows

The Regression Playground is the formal empirical-test page. It asks whether the CAR differences observed in the event study can be explained by comment-letter content features.

The backend runs 12 regression models:

Dimension	Choices
CAR windows	`CAR_-3_-1`, `CAR_-1_3`, `CAR_1_60`
Specifications	topic + severity; plus pre-event volatility; plus industry; plus both controls

The general formula is:

```
CAR_window ~ 0 + C(topic) + severity_score + optional controls
```

The `0 + C(topic)` design means all topic coefficients are displayed directly. There is no hidden omitted baseline topic. Each topic coefficient can be read as that topic's conditional CAR level, holding included controls fixed.

How to Read the Regression Table

Table element	Meaning
Coefficient	Estimated relationship between a variable and CAR
Parentheses	Robust standard error, not p-value
Stars	Statistical significance based on p-value
R-squared	Share of CAR variation explained by the model
Adjusted R-squared	R-squared adjusted for model complexity

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Interpretation Notes

- Topic coefficients compare topic-specific conditional CAR levels across SEC comment-letter types.
- `severity_score` measures whether more severe letters within topic groups are associated with higher or lower CAR.
- The four specifications show whether the topic/severity patterns are stable after adding pre-event volatility and industry controls.
- The immediate window, `CAR[-1,+3]`, is the main short-window market-reaction test; `CAR[-3,-1]` checks pre-event movement and `CAR[+1,+60]` checks post-event drift.

> Full 12-Model Coefficient Table

> Model Formulas and Diagnostics

> Regression-Ready Analysis Dataset

> Why Regression Matters

Regression Conclusions

- **Regression evidence is weaker than Topic CAR.** Best R-squared is 0.105; best adjusted R-squared is 0.046.
- **Severity is not supported.** `severity_score` is significant in 0/12 models, so H4 is not supported in this baseline.
- **Topic has partial evidence.** Topic coefficients are significant 13 times across the 12-model suite, giving limited exploratory support for H3.
- **Overall conclusion:** Regression does not fail the project; it shows that topic/severity have limited explanatory power for cross-sectional CAR in the current 100-company sample.

Main Numeric Results

Result	Value
Number of regression models	12
Severity significant models	0 / 12
Topic coefficient significant occurrences	13
Highest R-squared	0.105
Highest adjusted R-squared	0.046

Regression Interpretation

- Regression evidence is weaker than Topic CAR evidence.
- `severity_score` is not statistically supported in this baseline: it is significant in 0 out of 12 models.

- Topic has partial exploratory evidence: some topic coefficients are statistically significant, but the pattern is not strong enough for a broad causal claim.
- The highest adjusted R-squared is 0.046, so the models explain only a small share of cross-sectional CAR variation.

How to Explain This to Teammates

Regression answers a stricter question than the event study:

> Are the differences in CAR systematically explained by topic and severity after adding controls?

The current answer is modest. Topic appears more useful than severity, but the regression results are not strong enough to claim that topic and severity fully explain market reactions.

This does not mean the project failed. It means the event-study signal is more visually and descriptively clear than the cross-sectional regression evidence in the current 100-company baseline sample.

Combined Takeaway

Component	Role	Current conclusion
Topic CAR	Descriptive event-study evidence	Market reactions differ by topic
Regression Playground	Formal cross-sectional test	Topic has partial evidence; severity is not supported

The strongest presentation claim is:

> SEC comment-letter events show topic-specific market-reaction patterns. However, in the current baseline sample, topic and severity only partially explain cross-sectional CAR variation in formal regressions.

Suggested Presentation Framing

- Lead with Topic CAR because it is easier to understand visually and directly connects to the event-study design.
- Use Regression Playground as the formal robustness layer.
- Do not overclaim statistical significance. The regression results are useful because they show what the current evidence can and cannot support.
- State clearly that this is a baseline 100-company project version. A larger S&P 500 version may improve power and stability.

Source Notes

- Screenshots were taken from the local Streamlit dashboard at `http://localhost:8501/Final_Results`.
- Topic CAR values come from the project output table `outputs/tables/topic_car_summary.csv`.

- Regression results come from the 12-model backend regression suite in `outputs/tables/regression_coefficients_long.csv` and `outputs/tables/regression_model_summary.csv`.